

AgriShield

A Season-Long, Solar-Powered Companion for Smallholder Farmers — From the Field to a Fair Market

Field	Detail
Team / organisation name	AgriShield
Competition category	Tertiary Level
Primary sub-theme	AgriTech, Food Security and Farm-to-Market Innovation
Secondary alignment	Smart Communities, Green Technologies and Future Skills (solar power); Financial Inclusion and SME Market Access (market layer)
Province / location developed	Manicaland (Africa University, Mutare)
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This document follows the eleven-part structure and seven-page limit set out in Annexure 1 of the Expo Terms of Reference. Every technical and financial claim in it is either backed by a real, named source or clearly marked as a proposal not yet tested — the same honesty standard this team held itself to for its original Hack4Africa 2026 submission, of which this Expo entry is a continuation.

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3. Team Composition

Name	Role	Qualification / Focus	Contact
Mark Chindudzi	Technical Lead	4.1 Computer Science, Africa University	[to be provided]
Patience Mupikeni	Team Member	[to be provided]	[to be provided]

Both team members represent AgriShield at this Expo; the wider project was first built for the Hack4Africa 2026 CEAS hackathon at Africa University by a four-person team, of which Mark Chindudzi is a continuing member.

4. Solution Specification

4.1 Problem addressed

Smallholder farmers lose crops at three different points, and get a weaker price even for what survives. Post-harvest, food spoils in storage before it can be sold — the UN's Food and Agriculture Organization (FAO) reports that, worldwide, about one-fifth of the value of fruit and vegetables is lost after harvest, with Sub-Saharan Africa losing close to double the share lost in North America and Europe. Before harvest, a plant disease or a drying patch of land often goes unnoticed until it is too late to act. And even a good harvest often reaches a weaker buyer than it should, because there is no easy way for a farmer to know where the best price is, or to reach a buyer directly.

4.2 Solution objectives

- **Give an early, physics-based warning before stored food spoils**, using a real formula (spoilage roughly doubles for every 10°C rise in temperature) rather than a fixed shelf-life guess.
- **Screen sick plants from a photo, fully offline**, so a farmer with no signal still gets a fast first opinion.
- **Watch a whole farming district from free satellite imagery** to catch a drought or stress signal at a scale no single farmer could see alone.
- **Connect a farmer directly to a real buyer or a real input seller**, starting with a simple, low-automation contact introduction, before any scored marketplace.
- **Extend into the field with a multi-purpose Agri-Rover**, a solar-charged scouting robot adapted from this team's own Campus Compass navigation project (Section 10), able to carry a disease-screening camera, a mechanical weed trimmer and a seed/fertiliser dispenser — a genuine multi-year hardware roadmap goal, named honestly here as a direction, not a near-term build.

4.3 User and stakeholder needs

- **Smallholder farmers** need low-cost, offline-first tools that work on a basic phone with no data plan, and a plain-language answer, not a technical one.
- **Depot and cooperative operators** need early warning of spoilage risk across everything they store, before it becomes a loss, and a way to reach many farmers through one shared channel.
- **Agro-dealers, manufacturers and produce buyers** need a direct, low-cost channel to real, ready sellers, without needing their own field agents.

- **Extension officers, agronomy partners and regulators** need a tool that is honest about its own limits, defers a hard case to a real expert rather than pretending to replace one, and never states a figure it cannot back up.

4.4 Key functional and non-functional features

Type	Feature
Functional	Solar sensor box giving a live, temperature-adjusted shelf-life countdown and a separate mold-risk alert for stored grain.
Functional	Offline, on-device crop disease screening from a leaf photo, trained on the public PlantVillage dataset.
Functional	District-level satellite stress and soil-moisture mapping via free Sentinel-2 imagery (Google Earth Engine).
Functional	Weather, market-price and planting-window alerts, each pulled from a real free public data source.
Functional	A plain SMS/USSD/voice contact-introduction feature connecting a farmer's ready crop to a known buyer.
Non-functional	Works fully offline and syncs only when a connection returns; reaches any phone via text, USSD or a spoken call, with no smartphone required.

5. Target Market

The primary users are smallholder crop farmers in Zimbabwe, starting in Manicaland, and the collection points, depots and cooperatives they already sell through — reachable even where smartphones and reliable data are rare, because every alert also works as a plain text, a phone menu, or a spoken call. A secondary market is agro-dealers, input manufacturers and produce buyers who gain a direct, low-cost channel to real, ready sellers through the contact-introduction feature described in Section 4.4.

The value proposition is straightforward: a smallholder farmer using informal storage today has no early warning before spoilage, no offline way to check a sick plant, and no visibility into where a fair price actually is. AgriShield replaces guesswork with a real, physics- and data-based signal at each of those three points, at a hardware and running cost aimed squarely at what an individual farmer or a small cooperative can actually afford — not an enterprise price point.

Zimbabwe's own renewable-energy agency (IRENA) has already sized the loss-reduction opportunity at roughly \$128 million for solar cold storage alone, a market AgriShield's own sensor data is designed to build the evidence base for, rather than guess at (Section 7). The addressable base beyond that single figure is every smallholder farmer currently storing crops informally, without independent confirmation of a precise national count.

6. Business and Revenue Model

This is a proposed model, not yet tested with a paying customer — said plainly, in keeping with this project's own rule of never presenting a plan as more proven than it is.

Stream	Structure
Hardware	A low-cost sensor unit sold or leased to an individual farmer, a depot, or a cooperative, priced to stay near the cost of the components themselves (solar panel, battery, sensors, microcontroller).
Data and alert service	A small recurring fee for depots and cooperatives to access satellite, weather, and market-price alerts across their catchment area; a free basic tier for an individual farmer's own sensor alerts.
Market layer	A modest listing fee charged to agro-dealers, manufacturers, or buyers for a seller account, rather than a cut of any individual sale, matching the honest limit already named in Section 4.4 that AgriShield never touches the money in a farmer-to-buyer deal.
Grant and partnership funding	Cooperative, NGO, or government co-funding for initial hardware rollout to cooperatives — the model most comparable solar-agriculture hardware projects in this region rely on for their first phase.

Cost structure is dominated by the sensor hardware itself and the cloud/API costs of the satellite, weather and price feeds; marginal cost per additional farmer is low once the core software is built, since the same alert pipeline serves every user. Sales channel is expected to be direct engagement with farmer cooperatives, agricultural extension programmes and NGO partners, rather than individual retail sales, since a cooperative or depot is the natural first buyer for shared hardware and a shared data feed.

7. Design

AgriShield is built in layers, all reporting to one shared farmer record and one shared alert channel, so no module has to invent its own way of reaching a phone.

Layer	What it is
Sensor hardware	A solar-powered box with temperature, humidity and (for grain) CO2 sensors, plus the zeer pot-in-pot evaporative cooler — a real, century-old, no-electricity design that costs about one US dollar to build.
On-device software	A TensorFlow Lite model trained on the public PlantVillage leaf-photo dataset, running fully offline on a farmer's phone.
Satellite and data layer	Free Sentinel-2 imagery via Google Earth Engine for district-level stress mapping, plus free public weather (Open-Meteo) and vegetable-price (Zimbabwe's Agricultural Marketing Authority) feeds.
Market layer	A plain contact-introduction feature first (Section 4.4), pointing toward a fuller, explainable scored-matching engine later — the same explainable-AI design principle (compute a real answer first, then explain it, never guess) already proven in a related project by this team's wider network, Kuwana.
Delivery channel	Africa's Talking, a real, confirmed service carrying every alert as SMS, USSD, or a spoken voice call, so no smartphone or data plan is required to receive one.

User interface: for a farmer, the interface is the phone they already own — a text message, a simple numbered phone menu, or a spoken call — deliberately not a smartphone app, so that access never depends on owning one, and never excludes a farmer who cannot read, since a spoken call needs no literacy at all.

Component	Recommended technology	Why
Farmer-facing app	Flutter	One codebase for Android, the realistic majority device among smallholder farmers, with mature on-device support for the offline disease model.
Central server	Python (FastAPI) with a Postgres database	Only needed for what must be shared across many phones — satellite maps, price and weather data, synced records.
Satellite pipeline	Python, via Google Earth Engine	The same free tool already confirmed for the satellite layer, run automatically on a schedule.
Disease model	TensorFlow, trained on PlantVillage	Trained once centrally, then runs on the phone with no internet needed.
Delivery layer	Africa's Talking	Already confirmed to support text, USSD and voice calls across all of AgriShield's alerts.

8. Implementation

Development follows a deliberately staged plan, each stage reusing the same offline-first, one-farmer-record foundation:

- **Stage 1 — the sensor and the cooler.** The most physically real and demonstrable piece, tested against two live thermometers (inside and outside the cooler) to show a genuine, measured temperature difference.
- **Stage 2 — offline disease screening.** Trained and tested against PlantVillage's public photo set; a well-known 2016 follow-up study (Mohanty, Hughes and Salathé) is used honestly in this team's own documentation to flag that messy, real-field photos are harder than the dataset's own clean test photos — so this is always presented as a first opinion, never a final answer.
- **Stage 3 — the satellite layer.** Independent of the first two, built and shown in advance of any live demonstration, since satellite imagery cannot be generated live on demand.
- **Stage 4 — weather, price, planting and waste-to-feed modules.** Each added only once a real, checked public data source was confirmed for it — several were initially set aside for lacking one, and only reinstated once a genuine source was found.
- **Stage 5 — the market layer.** The simple contact-introduction version first, since it needs only a maintained buyer list and no scoring engine; the fuller scored version once real buyer and seller volume justifies it.

Testing and validation to date has been documentation-level and small-scale bench testing of the sensor and cooler pairing; a full field pilot with real farmers is the next validation step, honestly not yet completed.

8.1 Demonstration readiness

Stages 1 through 3 are demonstration-ready today: two live thermometers, inside and outside the zeer cooler, showing a real measured temperature gap; a phone that actually rings with a spoken shelf-life warning; an offline leaf-photo check answered in seconds with venue internet turned off; and a pre-built satellite stress map for a real district. Stages 4 and 5 (the weather, price, planting, waste-to-feed and market modules) are each grounded in a real, checked data source or working reference architecture, but are roadmap items, not yet built end-to-end — stated plainly here rather than implied as finished.

9. Social and Technological Impact

UN Goal	How AgriShield helps
Goal 2 — Zero Hunger	Less food is lost to spoilage or missed disease, so more of what is grown is actually eaten or sold.
Goal 1 — No Poverty	Farmers keep more of what they earn by losing less food and reaching a fairer price.
Goal 7 — Clean Energy	Every sensor and, in the roadmap, every future hardware piece runs on solar power, by design.
Goal 13 — Climate Action	Drought, weather and soil-stress warnings help farmers adapt to a less predictable climate.
Goal 9 — Innovation and Infrastructure	Builds simple, low-cost technology for places without steady power or internet, instead of assuming they have both.
Goal 12 — Responsible Consumption	Directly targets food grown but never eaten, and waste turned into livestock feed instead of thrown away.

Inclusion and accessibility: the whole system is designed around a basic phone and no data plan, not a smartphone and steady internet, so it reaches the farmers most exposed to these losses rather than only the already well-connected; a spoken voice call also means a farmer who cannot read, or a farmer with a visual impairment, is not excluded from any alert. Ethical stance: the disease screener is always presented as a first opinion, never a final answer, and every number in this document is either backed by a real, named source or clearly marked as an estimate still to be checked — including the business model in Section 6, which is a proposal, not a tested result.

10. Previous Projects

AgriShield itself was first built and submitted for the Hack4Africa 2026 CEAS hackathon at Africa University, where its sensor, disease-screening and satellite modules were designed, sourced and, for the sensor and cooler, bench-tested. Two further pieces of evidence, from adjacent projects this team's own members have built, ground specific parts of this Expo submission rather than leaving them as unproven ideas:

- **Campus Compass (Kinetic Directory).** A real, demoed navigation robot built at Africa University — a Raspberry Pi 4 and Arduino pair, A* pathfinding, ultrasonic and LiDAR obstacle avoidance — whose listed contributors include a member of this team. It is the working starting point for AgriShield's longer-term field-robot concept.
- **Kuwana.** A real, working explainable-AI comparison platform built by this team's wider network, whose "compute a real answer, then explain it" design principle grounds AgriShield's own market-layer approach, and whose architecture has since been proven a second time in a sibling product (ZivaBasa), strengthening confidence that the same pattern is genuinely reusable.

Full documentation, source citations and build artefacts for all of the above are held in this team's own project records and are available on request.

11. User / Product Manual

- **1. Set up.** The solar sensor box is placed where food is stored; no wiring or mains power is needed. A farmer's name, rough location and crop are registered once, by the farmer or a depot operator, on any phone.
- **2. Receive alerts automatically.** A text, a phone-menu message, or a spoken call arrives when shelf-life is running low, mold risk is rising, a satellite signal shows district-level stress, rain is expected, or a price or planting window changes — no app to open or check.
- **3. Check a sick plant.** A leaf photo taken on any phone with the AgriShield app gives an offline first opinion in seconds, with a note to see a real extension officer for a hard or unclear case.
- **4. Get connected to a buyer.** When a batch is ready, a farmer may receive a message naming a real, known buyer and their contact details; the farmer calls and negotiates directly — AgriShield makes the introduction, not the deal.

Full technical setup for a depot or cooperative deploying multiple sensor units is intended to be documented in a deployment guide alongside the working build, following the same plain-language standard as this document.